

MOVING FORWARD WITH BUILDING PERFORMANCE

BY COURTNEY MORIARTA

“**W**hat you’re doing is going to make all of our work worthwhile,” a colleague said to me as I packed my things up following a recent presentation. This vote of confidence came from a highly respected individual who has invested years of effort into building science research and the development of standards for the building science community. I was pleasantly surprised by the statement. The Building Performance Institute (BPI), the nonprofit that I direct, promotes excellence in the contracting trades by establishing standards of performance and technician certification. But the impacts of our work extend far beyond the contractors we certify and the customers they serve. Our work may well permanently change the face of the building science industry and its connection to the consuming public.

This is good news for the pioneers of our fledgling industry. BPI’s commitment to raising the bar both technically and professionally for the contracting, building, and weatherization industries by developing building science-based certification processes for the building trades offers potential benefits to many audiences (see “BPI Certifications” and “Testimonials to Certification”). Certified contractors benefit through increased sales and decreased liability. Consumers benefit from high-quality installations and safe, affordable living environments. But beyond that, BPI’s promotion of building science-based certifications puts into action—and justifies—two decades of building science research and development. If you’ve ever asked yourself, “Why are we doing this?” the answer may be about to reveal itself in spades.

It was not so long ago that the building science community was viewed as a fringe minority. Plagued by disasters and near-disasters caused by suboptimal technology and overzealous attempts at making homes more energy efficient, we energy people have sometimes been regarded as the source of more problems than solutions. We are all too familiar with the reputation earned by energy programs that weatherized houses without controlling moisture and ventilation properly. How many among us had the moral fortitude to put up with the poor light quality and the 60 Hz flicker of magnetic-ballast CFLs in our own homes as we forced them down the throats of our clients? And who can forget the solar water heating debacle, which sent even

the strongest proponents of solar power running for cover? Showerheads that mist more than they rinse, urea-formaldehyde foam, and Jimmy Carter’s advice to the country to “put on an extra sweater”—the list goes on.

Having survived those pioneering attempts at “fixing” homes and saving energy, and having learned a few lessons in the meantime, we are a much better, stronger, and smarter industry. We know that testing in and testing out is a good way to make sure that a house functions the way we expect it to. We understand that any house is a system of interactive components, and that we are asking for trouble if we choose to ignore that simple fact. New technologies have emerged, resulting in vastly improved building products,



Tony Peters (BPI) and Edeson Forbes (Mt. Hope Housing Corporation, Bronx) inspect a rooftop ventilation fan.

MAURIE HEGGEN



Bill Putzer (Comfort Windows, Syracuse, New York) measures carbon monoxide production in an oven.

materials, and equipment. And maybe best of all, an assortment of reliable performance-testing techniques and equipment—blower doors, CO detectors, infrared cameras, among others—is out there just waiting to become part

of the standard package of tools of the trade for practitioners.

Slowly but surely, programs are similarly evolving over time, adopting improved standards and requiring a wider skill set among the technicians who work

for them. In recent years, new-construction programs have led the way by introducing building science-based approaches to the private market. Having broken through the public/private sector barriers, we find large-production builders and product manufacturers now supporting, and even embracing, programs like *Environments for Living* and *Building America*. The success of these partnerships is paving the way for retrofit initiatives like EPA's *Home Performance with Energy Star* to take hold in the private sector as well, and to manifest long-term change in the contracting marketplace.

The Brave New World for Building Performance

The brave new world we now face offers both potential and challenge. Can market transformation become reality? Will a mature industry emerge from what was once considered peripheral to the reality of the private market? The indicators are positive for both questions.

BPI CERTIFICATIONS

BPI provides certifications for the following job titles:

Building Technician I (BT-I). This is an installer level certification that is primarily used by weatherization assistance programs for air sealing and insulation crew members. This certification is available to all contractors but is not required for corporate accreditation.

Heating Technician I (HT-I). This is an installer level certification that is primarily used by weatherization assistance programs for their heating service technicians. This certification is available to all contractors but is not required for corporate accreditation.

Building Analyst I (BA-I). This is the first level of certification required for technicians who perform whole-house audits and/or intend to earn any Specialist level certification. BA-I covers the fundamentals of building science

and whole-house diagnostics and is a prerequisite for all Specialist level designations. This job title is suitable for an auditor, inspector, or HERS Rater.

Shell Specialist. This is a supervisory level certification for contractors performing building envelope related work (such as windows, insulation, and air sealing).

Heating Specialist. This is a supervisory level certification for contractors performing heating system related work (such as tune-ups, repairs, replacements, and installations).

Cooling Specialist. This is a supervisory level certification for contractors performing work related to refrigerant-based heating and/or cooling systems (such as tune-ups, repairs, replacements, and installations).

Mobile Homes Specialist. This is a supervisory level certification for

contractors performing retrofit work on mobile homes for the purpose of energy improvement.

Multifamily Building Analyst. This job designation is for analysts and engineers who conduct comprehensive energy audits on multifamily buildings. The focus is primarily on larger buildings, but the principles may apply to any multifamily building.

Multifamily Building Operator. This job designation is for personnel responsible for the efficient and safe operation of multifamily buildings. Building superintendents and/or building managers may benefit from this certification.

For detailed descriptions of any of the job designations listed above, contact BPI and ask for a Skill Set cut-sheet describing the certification you are interested in.

New York contractors, introduced to building performance with the assistance of incentives provided by the New York State Energy Research and Development Authority (NYSERDA), report that their businesses have developed in ways they never knew were possible. Some contractors report that their businesses have doubled in size since they took on the challenge of building performance as a business enterprise.

The implications for trades contractors do not stop at the obvious modifications to their technical approach to jobs. For a private market contractor, successfully adopting a building performance approach requires a complete cultural change to the contracting firm. Moving beyond a simple trades approach requires contracting firms to make corporate level changes, ranging from the establishment of dedicated administrative staff to instituting job-tracking and financial management programs capable of keeping track of partially completed jobs. While these business systems are commonplace among builders and general contractors, they are not typically found in any but the largest of HVAC companies and are virtually nonexistent in insulation firms.

So why do they do it? We find that once these contractors take the time to go through a class and learn how buildings really work, we have them hooked. They arrive as skeptics and leave with a new perspective on their work. By completing the certification and corporate accreditation process, the newly converted are equipped with the knowledge, tools, and technical standards they need to begin delivering whole-house services. Since we know that a few weeks of training and passing a competency exam are insufficient to prepare technicians for what they will find in the real world, technical support is available through BPI and its Technical Committee.

Plotting a Course

We now have a mechanism for bringing the standards and procedures

TESTIMONIALS TO CERTIFICATION

BPI has certified technicians in 22 states. From time to time, our contractors and the customers they serve send us testimonials about their experiences and tell us how BPI has helped them. Here's a sampling of what they have to say:

Since becoming BPI certified, I have had several CO readings in the thousands of parts per million and have promptly shut down the system, recommended immediate service, and felt confident that I had helped avert a potential hazard. On one occasion, I instructed the customer to turn up the thermostat while I made my way downstairs to do a combustion efficiency test. Well, just as I approached the boiler it fired up, and within 15 seconds I was running for the stairs coughing and choking. The chimney was completely blocked up with carbon deposits and was also shared with a gas water heater! What I learned from this experience was how quickly things can turn dangerous out there, and that we are doing a great service for our customers. Keep up the good work and stay alert.

—Jon Patton, BPI-certified building analyst 1, Honeywell DMC, Saugus, Massachusetts

The Cape Light Compact and its low-income energy efficiency program vendor, Housing Assistance Corporation (HAC), had a recent experience that demonstrates the importance of instrumented home energy audits. Staff went through a course of training to prepare for the BPI written and field tests. A home was made available to help these future BPI-certified technicians prepare for the field performance exam at the Building Analyst I level. Everyone got oriented to the gas-fired DHW and boiler in the basement of the home selected and observed the nearby woodshop and sawdust on the floor as the health and safety checks began. The spillage, vent, and CO on the DHW proceeded smoothly and the testing moved over to the boiler that had sat idle during the summer. Upon firing up the boiler, the CO flue gas steadily climbed over 100 ppm, then

200 ppm, and as the increasing trend was not letting up, the steps to shut down, "red-tag," evacuate, and call for service were quickly followed. In addition, during the outdoor inspection, a gas leak was also detected that resulted in a call and service performed by the local gas utility. Needless to say, the owners of the test home are pleased with the BPI testing that was performed, as well as the direct experience these future technicians gained in the field. Finally, remind those customers with woodshops near their boilers to clean up the sawdust—it may save their lives!

—Kevin Galligan, Energy Efficiency Program Manager, Cape Light Compact, Barnstable, Massachusetts

In August 2003, I received a call from a gentleman saying he built his home himself in 1960, had heard about the Home Performance Program and BPI, and wanted an energy audit done. He said he just wanted to know why his kitchen was cold. When I started the inspection, he said he thought his wife was "nuts" because she kept feeling a draft by the kitchen cabinets, which he swore up and down he and his brothers had built and insulated themselves. I turned on the blower door and the cabinets leaked like a sieve! The attic insulation was installed well, but I asked if I could crawl up and take a better look. He insisted I would not find anything because he "did it personally." I said I would feel better if I just had a look, so he said, "Be my guest." I crawled over to the top of the kitchen and lifted the fiberglass insulation and looked straight down into the kitchen soffits! They never covered the soffits when they built the home; you could see right to the cabinets. This man could not believe it! He was most impressed with the blower door and could not believe all the air coming from around the cabinets. He said, "Please don't tell my wife. I've been telling her she's crazy for the last 40 years!"

—Michelle Knaszak, BPI-certified building analyst 1, shell specialist, heating specialist, cooling specialist, Almac Energy, Hamburg, New York

developed from two decades of building science research to trades contractors whose primary business is construction and home improvement. For now, successful building performance contracting businesses are being created by visionary contractors who are willing to move on to the next level. Some contractors are making this transformation independently; in areas that receive program support from local public sector agencies or private market sponsors, more contractors are choosing to transform their businesses.

And a transformation it is. Contractors who have successfully adopted a whole-house approach, based on BPI's standards, report that they will not go back to the old way of doing business if and when the program funding goes away. The incentives made it possible for them to gear up for change, but the changes we have brought to them stand on their

merits alone. So my friend was right; everything we energy people have been working so hard for is finally coming full circle. By focusing our



efforts on getting contractors to apply building science principles to their work, we are really making buildings better—which was the point all along.

We at BPI are working to take the building performance movement to

the next level. But we are not doing this alone. It is only through the cooperative efforts of our partnerships with technical experts, trainers, program sponsors, government, and the building science industry at large that we can realistically effect change of any significance. To those who came before us, we are grateful for your experience and willingness to share your knowledge and support this movement.

Are we changing the world? Yes, I think we are.



Courtney Moriarta, executive director of BPI since 2000, trained as an engineer and is responsible for fixing more than a few buildings. She believes that change in the world is inevitable, and that we are responsible for guiding that change in the right direction.

BPI